

Max Elinson

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EDUCATION

University of Toronto <i>HBSc in Statistics, Computer Science, and Mathematics</i>	Toronto, ON <i>June 2026</i>
• Software Design, Databases, Machine Learning, Statistics and Data Analysis, Mathematical Modeling	

RELEVANT WORK EXPERIENCE

Department of Fisheries and Oceans <i>Data Solutions Engineer</i>	Nanaimo, BC <i>May 2025 - Present</i>
• Built Python and R ETL pipelines to ingest, validate, and publish survey, collection, and lab datasets for scientists, analysts, and regional managers.	
• Produced official maps, figures, and dashboards supporting Pacific salmon stock assessment, management planning, and internal/external data requests.	
• Analyzed 40 years of salmon migration data to identify sampling optimizations projected to reduce survey effort by 10% without reducing predictive precision.	

Ninepoint Partners LP <i>Software Engineering Intern</i>	Toronto, ON <i>May 2023 - August 2023</i>
• Built a full-stack web application with a C#/.NET Core backend, JavaScript/HTML frontend, and Umbraco CMS integration.	
• Developed tax calculators and external API data pipelines that improved user engagement by 10% and supported analytics workflows.	
• Implemented automated email routing that reduced response times by 30%, plus reusable HTML/CSS components for faster page development.	

NYU Langone Health <i>Student Research Intern</i>	New York, NY <i>May 2022 - June 2022</i>
• Built R modules to simulate research study data using dummy variables, Markov chains, and spline curves.	
• Refactored core and utility functions to improve runtime efficiency by 20%.	

PROJECTS

Toronto Development Timeline Map [\[link\]](#), *April 2026*

- Built an interactive 3D MapLibre/deck.gl timeline of Toronto development applications using 2025 massing, 2015 before-state geometry, planning records, and PMTiles vector layers.
- Designed a Python geospatial pipeline to join applications with building footprints, classify project status, generate as-built layers, and audit temporal building visibility.

Toronto Traffic Heatmap [\[link\]](#), *April 2026*

- Built a browser-based traffic visualization from Toronto turning movement count data, rendering cars, transit, pedestrians, and bikes as animated particles and road-density layers.
- Processed TMC observations with an OpenStreetMap road graph, generated directed segment flows and hourly route pools, and added Web Worker loading for smoother interaction.

Spatial SIR Model with Individual Mobility [\[link\]](#), *March 2025*

- Built a hub-based stochastic SIR model with individual movement represented as a Markov chain.
- Used state-space lumping and Monte Carlo simulation in Python to quantify outbreak-size distributions and identify high-risk hubs.

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, JavaScript, HTML/CSS, C#, Java, C++, C
- **Technologies:** Pandas, GeoPandas, MapLibre GL, PostgreSQL, SQLite, ArcGIS, PowerBI, Django, .NET Core, Git, L^AT_EX

VOLUNTEER ACTIVITIES

Scatterbrain Magazine <i>Vice President, Head of Website Design</i>	Montreal, QC <i>May 2024 - Present</i>
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